Manufacturing Cost and Profitability Analysis

Complete worked solution

For each product batch, compute material cost from units and material cost per unit, then add labour, machine cost, and other direct charges. Wastage is a percentage of material cost. Unit production cost is total cost divided by saleable units. Profit is selling revenue minus total cost; report profit margin and whether the batch meets the requested target margin. Reject invalid quantities, negative costs, and wastage over 100%.

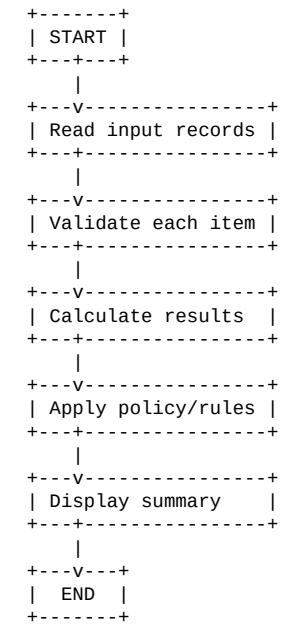
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read batch input and target profit margin
  Validate inputs and saleable quantity
  Calculate material cost and wastage cost
  Add labour, machine, and other costs
  Calculate unit cost, revenue, profit, and margin
  Compare actual margin with target
  Display cost breakdown and target result
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
# (batch, units, material/unit, wastage_percent, labour, machine, other, sale_price/unit)
batches = [
    ("B-11", 500, 120, 4, 18000, 12000, 5000, 190),
    ("B-12", 250, 90, 7, 14000, 9000, 2500, 145),
]
target_margin_percent = 15
for batch, units, material_unit, wastage_pct, labour, machine, other, sale_price in batches:
```

```
numbers = [units, material_unit, wastage_pct, labour, machine, other, sale_price]
valid = True
for number in numbers:
    if number < 0:
        valid = False
if units <= 0 or wastage_pct > 100 or not valid:
    print(batch, "invalid batch data")
    continue
material_cost = units * material_unit
wastage_cost = material_cost * wastage_pct / 100
total_cost = material_cost + wastage_cost + labour + machine + other
revenue = units * sale_price
profit = revenue - total_cost
margin_percent = profit * 100 / revenue if revenue > 0 else 0
unit_cost = total_cost / units
meets_target = margin_percent >= target_margin_percent
print("Batch:", batch)
print("Material:", round(material_cost, 2),
      "Wastage:", round(wastage_cost, 2),
      "Labour:", labour, "Machine:", machine, "Other:", other)
print("Total production cost:", round(total_cost, 2),
      "Cost per unit:", round(unit_cost, 2))
print("Revenue:", round(revenue, 2), "Profit:", round(profit, 2),
      "Margin %:", round(margin_percent, 2))
if meets_target:
    print("TARGET MET")
else:
    print("TARGET NOT MET")
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.